

**Zhiyuan Xiao** +1 585-660-9096 |zxiao17@u.rochester.edu| Rochester, NY, USA

EDUCATION

University of Rochester

Rochester, USA

Bachelor of Science, Computer Science; Bachelor of Arts, Statistics

Sep 2023 - Dec 2026

Overall GPA: 3.94/4.00; GRE 327

Technical skills:

Languages: Python, C, Java, R, SQL, Swift

Computer Vision and Machine Learning: PyTorch, OpenCV, NumPy, CLIP, T2I-Adapter, GPU-accelerated training and inference, Image/Video Generation, DINO

Systems and Tools: Linux, Docker, LaTeX

PROFESSIONAL EXPERIENCE

Standard Robotics

Shenzhen, China

Embedded System Engineer Intern

Jun 2026 - Jul 2026

- Developed and deployed a CAN/LoRa firmware upgrade for remote power control across 150 devices, reducing start/stop command completion time from 600 to 180 seconds (70%) and increasing single-command success from 75% to 100%.
- Built 14 automated tests and performed 1 manual field validation covering firmware flashing, functional modules, and communication terminals, identifying and fixing a factory-floor issue.

Aurora: Agentic Video Editing Data and Evaluation

Nov 2025 - Present

Research Assistant in Prof. Jiebo Luo’s team at the Department of Computer Science, U of R

- Curated tens of thousands of image and video editing samples across eight operations and five formats, including subject-to-video (S2V) and SV2V.
- Developed a multi-stage quality-control pipeline using DINO, CLIP, and VLM filtering, removing more than 60% of low-quality samples and supporting the dataset's open-source release.
- Built Docker GPU benchmark environments on Linux to ensure reproducible results across library versions and driver configurations.

Deepano Technology, a leading provider of visual intelligence platform technology

Shanghai, China

Embedded System Engineer Intern

May 2025 - Aug 2025

- Optimized image-format conversion, denoising, and color-correction stages for embedded vision devices on the Horizon Robotics Journey 5 (X5) platform.
- Applied SIMD optimization and target-hardware profiling to increase processing throughput from 20 to 60 FPS (3x), while tuning algorithm parameters and software-hardware interfaces for runtime stability.

Definesys.com, China’s leader in Low-code iPaaS platforms

Shanghai, China

Java Architect Intern

Jun 2024 - Jul 2024

- Developed and maintained ETL tools for cloud data workflows and validated four localized iPaaS releases across 20 functions for functionality, performance, and reliability.

Publications

Yongsheng Yu, Ziyun Zeng, **Zhiyuan Xiao**, Zhenghong Zhou, Hang Hua, Wei Xiong, and Jiebo Luo. “Aurora: Unified Video Editing with a Tool-Using Agent.” *arXiv preprint. arXiv:2605.18748*, 2026. <https://arxiv.org/abs/2605.18748>

## ACADEMIC PROJECTS

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### **Multimodal Detection of AI-Generated Twitter Posts**, Python, CLIP, PyTorch

*Oct 2025 - Dec 2025*

- Developed a CLIP-based multimodal detection system for analyzing AI-generated Twitter posts by modeling image-text semantic alignment, and curated a Twitter multimodal dataset with systematic ablation studies on human vs. AI-generated pairs.
- Authored a 10-page research paper presenting the dataset construction, CLIP-based methodology, experimental evaluation, and analysis of multimodal generation artifacts.

### **AI-Generated Twitter Detection**, Python, PyTorch, OpenCV, NumPy

*Apr 2025 - May 2025*

- Developed a two-stage AI system (Sketch2Anime) that transformed simple hand-drawn line sketches into high-quality anime-style illustrations while preserving the artist's original vision.
- Implemented optimized Canny edge detection with a fine-tuned T2I-Adapter model that successfully handled backgrounds, diverse poses, and 5 types of art styles.

### **Dynamic Storage Allocator (Malloc Lab)**, C

*Apr 2025 - May 2025*

- Implemented a custom malloc/free/realloc package in C, ensuring 8-byte alignment, correct heap invariants, and libc-consistent semantics.
- Improved space utilization and throughput via free-block coalescing and free-list allocation, with a heap consistency checker for correctness validation.

### **Statistical Modeling Inference Project**, R

*Mar 2024 - May 2024*

- Analyzed BRFSS obesity prevalence across U.S. states (2013–2022) and quantified group differences using ANOVA (incl. Dunnett) and Kruskal–Wallis tests across age, race/ethnicity, education, and gender.
- Built a log-linear regression model with stepwise selection to estimate obesity rates and summarized annual trends, including post-2020 increase.